

Bowen Truelove

2326 Red Barn Rd, Christiana, TN 37037, United States | botruelove2002@gmail.com | [6159563796](tel:6159563796)

Profile

Systems Engineer with a strong software and automation focus, experienced in designing and supporting data-driven systems in manufacturing environments. Proven ability to understand complex, interconnected systems, break them into manageable components, and build reliable solutions using Python and SQL. Experienced in MES deployments, ETL pipeline development, and internal tooling, with a collaborative, documentation-driven approach to engineering and continuous improvement.

Accomplishments

- Played a key role in a **company-wide manufacturing retooling initiative**, supporting the successful deployment of a new **MES platform** and launch of a **new product line**.
- Designed and developed an **ETL pipeline** to consolidate CSV data from multiple IPC-controlled machines into a **SQL Server database**, enabling process engineers to make **data-driven decisions** and improve manufacturing KPIs.
- Created an **internal change management web application ("4M")** to standardize and track process, material, machine, and manpower changes, improving traceability and cross-team communication.
- Built a **Python-based automation framework** with structured logging, automated email alerts, reporting, and file handling to replace manual and error-prone operational tasks.
- Improved system reliability and reduced manual intervention by introducing **maintainable automation and data pipelines** aligned with production needs.
- Acted as a technical bridge between **operations, engineering, and vendors**, translating operational requirements into actionable technical solutions and system improvements.

Professional Experience

System Engineer 2, AESC US, Smyrna, TN

May 2025 — Present

- Own the operation, maintenance, and improvement of production systems in a **complex manufacturing environment**.
- Collaborate cross-functionally with operations, engineering, and vendors to **ensure system stability and alignment with production needs**.
- Support ETL pipelines, automation scripts, and internal tooling for **data collection, reporting, and process efficiency**.
- Document workflows, processes, and technical solutions to ensure **knowledge sharing and continuity**.

System Engineer 1, AESC US, Smyrna, TN

May 2024 — May 2025

- Assisted with **MES installation, configuration, and startup**, providing troubleshooting support during early production runs.
- Supported production operations through task automation, file handling, and data validation workflows.
- Developed and maintained lightweight internal tools to **improve process standardization and operational visibility**.
- Coordinated with multiple teams to identify system issues, escalate defects, and support feature requests.

System Engineering Intern, AESC US, Smyrna, TN

Feb 2023 — May 2024

- Provided first-line technical support for production systems, including legacy MES troubleshooting, IPC restoration, and printer/workstation support.
- Programmed and maintained industrial radios for **shop-floor communication and equipment integration**.
- Executed SQL-based data updates under supervision to ensure **accurate production records**.
- Gained hands-on exposure to manufacturing systems, MES environments, and production workflows.

Education

Middle Tennessee State University, Murfreesboro, TN

Bachelor of Science, Computer Science, Aug 2021 — Dec 2024

Motlow State Community College, Smyrna, TN

Associate of Applied Science, Mechatronics, Aug 2020 — May 2021

Technical Skills

Systems and Engineering	Understanding complex systems, Understanding system boundaries, interfaces, and data flows, Root-cause analysis/Troubleshooting, Manufacturing Systems (MES, Delmia Apriso)
Programming/Scripting	Python (automation, ETL pipelines, tooling, data processing) , SQL (data querying, transformations, reporting) , C#/ .Net (web development, EF Core) , C++/C (academic experience, foundational knowledge) , Swift (personal projects, active learning)
Communication	Explaining complex topics to technical and non-technical stakeholders, Visual communication using diagrams and system flow representations, Cross-functional collaboration with engineering, operations, and manufacturing teams